On Groups

$$G_1 \hat{=} G_2$$

Definition: Two Groups $(G, \cdot)$ and $(H, \circ)$ are said to be isomorphic if there is a bijection

$$\varphi : G \longrightarrow H$$

such that $\varphi(xy) = \varphi(x)\,\varphi(y) \quad \forall x, y \in G$.

φ is called an isomorphism.

ex.: $(\mathbb{Z}_2, +)$, $(\{\pm 1\}, \cdot)$

$$\varphi : \mathbb{Z}_2 \longrightarrow \{\pm 1\}, \quad \begin{aligned} 0 &\mapsto 1 \\ 1 &\mapsto -1 \end{aligned}$$

Check
$$\begin{aligned}
\varphi(0+0) &= \varphi(0) = 1 = 1 \cdot 1 = \varphi(0) \cdot \varphi(0) \\
\varphi(0+1) &= \varphi(1) = -1 = 1 \cdot (-1) = \varphi(0) \cdot \varphi(1) \\
\varphi(1+1) &= \varphi(0) = 1 = (-1) \cdot (-1) = \varphi(1) \cdot \varphi(1)
\end{aligned}$$

$\Rightarrow \mathbb{Z}_2 \hat{=} \{\pm 1\}$

ex. $(\mathbb{R}, +) \hat{=} (\mathbb{R}^+, \cdot)$

$\varphi : \mathbb{R} \longrightarrow \mathbb{R}^+, \quad \varphi(x) = e^x$

injective: $\varphi(x) = \varphi(y) \Rightarrow e^x = e^y \Rightarrow x = y$ ✓

surjective: Suppose $y \in \mathbb{R}^+ \Rightarrow \ln(y) \in \mathbb{R}$

Set $x := \ln(y) \rightarrow \varphi(x) = e^x = e^{\ln(y)} = y$ ✓

$$\varphi(x+y) = e^{x+y} = e^x \cdot e^y = \varphi(x)\cdot\varphi(y).$$

ex.: $U(5) \cong U(10)$

$$U(5) = \{1, \underset{3^3}{2}, \underset{3^1}{3}, \underset{3^2}{4}\} = \langle 3 \rangle$$

$$U(10) = \{1, \underset{7^3}{3}, \underset{7^1}{7}, \underset{7^2}{9}\} = \langle 7 \rangle$$

$$\varphi: U(5) \to U(10), \quad 3^k \mapsto 7^k$$

$1 \mapsto 1$
$3 \mapsto 7$
$4 \mapsto 9$
$2 \mapsto 3$

$k, l \in \mathbb{Z}_4$

$$\varphi(3^k \cdot 3^l) = \varphi(3^{k+l}) = 7^{k+l} = 7^k \cdot 7^l = \varphi(3^k)\cdot\varphi(7^l)$$

Definition: We say G is cyclic and generated by $g \in G$ if
$$G = \langle g \rangle = \{g^n \mid n \in \mathbb{Z}\}.$$

Theorem: If $G = \langle g \rangle$ with $|G| = \infty$ then $G \cong \mathbb{Z}$.

proof: Consider $\varphi: \mathbb{Z} \to G$ defined by $\varphi(n) = g^n$.

note: $\varphi(m+n) = g^{m+n} = g^m \cdot g^n = \varphi(m)\cdot\varphi(n)$

injective: Suppose $\varphi(m) = \varphi(n) \Rightarrow g^m = g^n$.

OEdA $m \geq n \Rightarrow g^{m-n} = e$

case 1: $m = n \Rightarrow \varphi$ injective

case 2: $m > n$. Set $k = m - n$

$$\Rightarrow \langle g \rangle = \{e, g, g^2, \dots, g^{k-1}, \underset{}{\overset{e}{g^k}}\}$$

$$\Rightarrow |G| = k < \infty \quad ⚡$$

So φ is injective. Now recall $\varphi: \mathbb{Z} \to G$,

surjective: Suppose $x \in G = \langle g \rangle \Rightarrow x = g^n$ for some $n \in \mathbb{Z}$. $\Rightarrow \varphi(n) = g^n = x$. □

Theorem: Suppose $G = \langle g \rangle$ with $|G| = n$. Then $G \cong \mathbb{Z}_n$.

proof: $\varphi: \mathbb{Z}_n \to G$ defined by $\varphi(m) = g^m$ $0 \leq m \leq n-1$

Well defined: Suppose $m \equiv m' \mod n$

$\Rightarrow n \mid m - m' \Rightarrow m - m' = k \cdot n \quad k \in \mathbb{Z}$

$\Rightarrow \varphi(m - m') = \varphi(kn) \Rightarrow g^{m-m'} = (g^n)^k = e$

$\Rightarrow g^m = g^{m'} \Rightarrow \varphi(m) = \varphi(m')$

Group law: Suppose $l, m \in \mathbb{Z}_n$

$$\varphi(l+m) = g^{l+m} = g^l \cdot g^m = \varphi(l)\cdot\varphi(m)$$

<u>surjectivity</u>: Suppose $x \in G = \langle g \rangle$

$\Rightarrow x = g^m$ for $0 \leq m \leq n-1$

$\Rightarrow \varphi(m) = g^m = x$

<u>injectivity</u>: Suppose $l, m \in \mathbb{Z}_n = \{0, 1, \dots, n-1\}$

$\varphi(l) = \varphi(m) \Rightarrow g^l = g^m \Rightarrow g^{l-m} = e$

case $l \neq m$: $\Rightarrow l - m = \{1, \dots, n-1\}$

$\Rightarrow g^{l-m} = e \Rightarrow |g| < n$

$\Rightarrow |\langle g \rangle| < n$ ↯

$\Rightarrow l = m$, hence injective

<u>Summary</u>

All infinite cyclic groups $\hat{=} \mathbb{Z}$

All finite cyclic groups $\hat{=} \mathbb{Z}_n$ for some n.